Session 1: Foundations of Marine Macroecology and Niche Theory

Modeling the distribution of biodiversity under global climate change

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Macroecology and global biodiversity patterns

Macroecology: the big picture

A statistical approach **focusing on large-scale ecological patterns** (not a separate field of Ecology; coined ~30 years ago).

Explores **relationships between biodiversity patterns** (e.g., species abundance levels, and the distribution of genes, species, ecosystems) **and the environment**.

Seeks the principles governing ecological systems, ignoring fine-scaled, localized details.

e.g.,

What environmental variables drive the distribution of a mangrove species?

How global climate change may impact kelp forests biodiversity?

Macroecology: the big picture

Compares different **populations** (i.e., different locations), **species** and **ecosystems**, across large areas and long-time periods, **to generalize and scale up results** - ultimately to figure out the global distribution drivers and patterns of biodiversity.

Contrasts with localized, reductionist experimental approaches that may struggle to generate broad predictive theory.

Global Biodiversity Patterns: the non-randomness of life

Species aren't found everywhere – their distributions are limited.

Macroecology helps understand these limits, which are mostly set by:

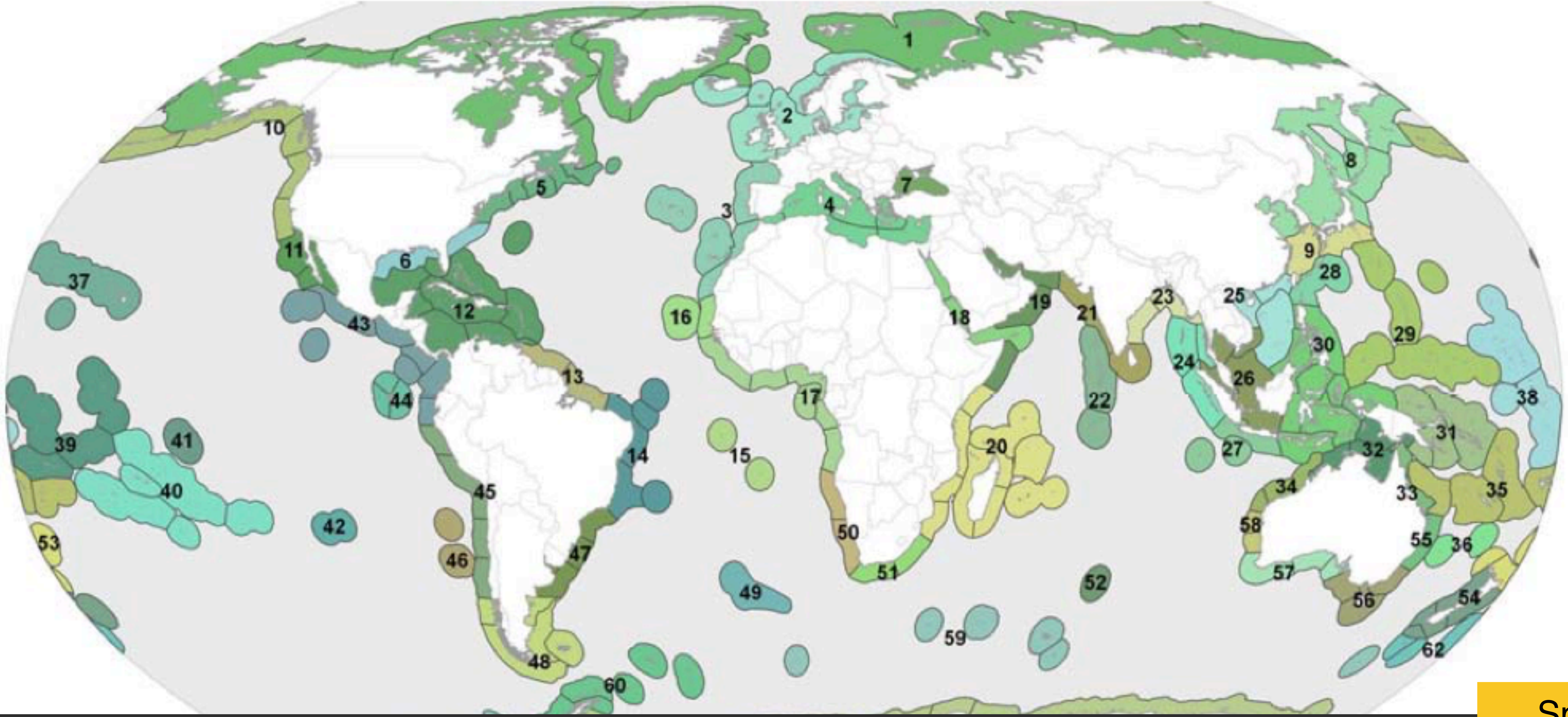
- **Niche requirements:** e.g., environmental tolerances and resource needs.
- **Dispersal constraints:** e.g., the ability to reach and establish populations in suitable areas.

Without these constraints, species could potentially occur everywhere, leading to absent or random spatial patterns in distribution and diversity.

Species distributions shape well defined biogeographic patterns

Classification systems based on **species assemblages and levels of endemism** show sharp biogeographic regions - within regions similar biodiversity / between regions dissimilar biodiversity.

e.g., *Marine Ecoregions of the World; nested system (12 realms, 62 provinces, and 232 ecoregions).*

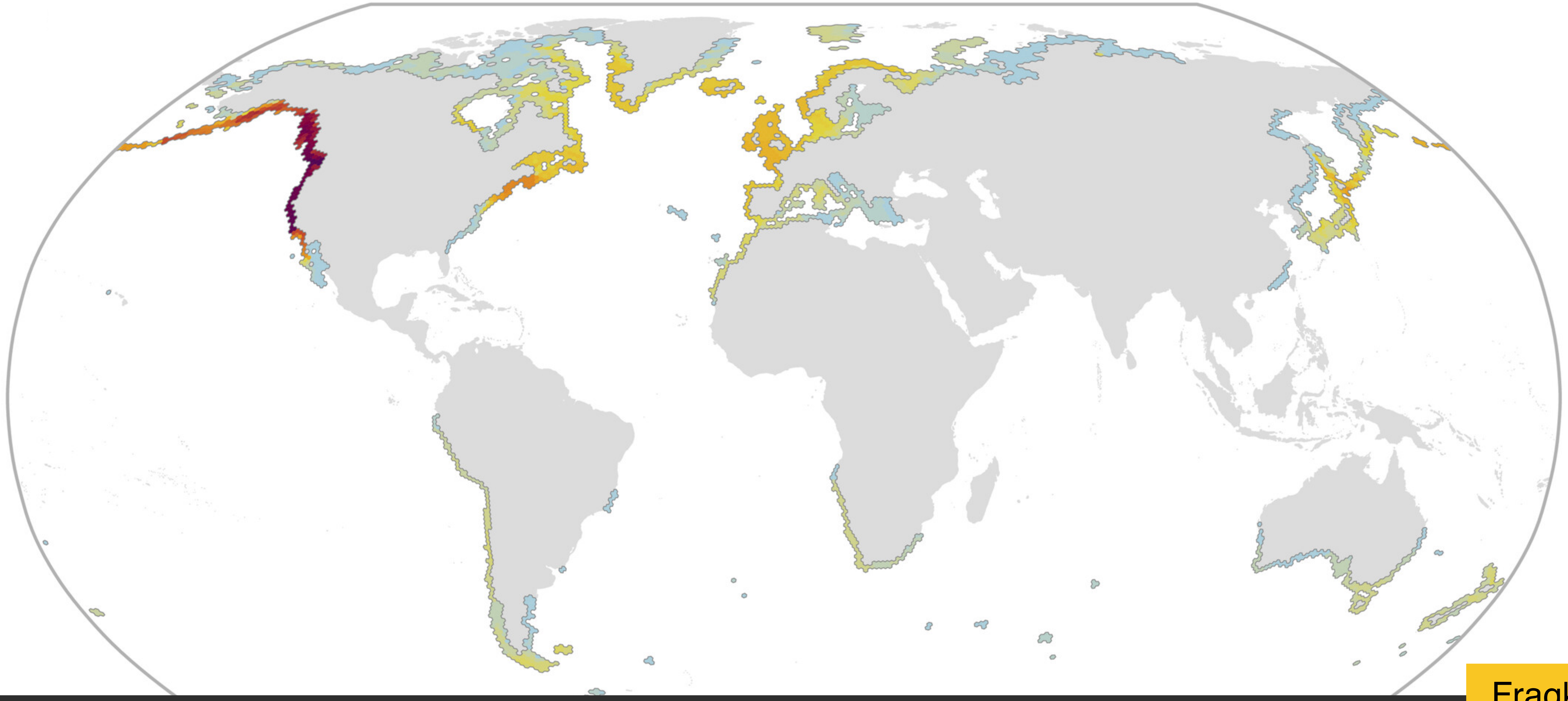


Spalding et al., 2007

Species distributions shape well defined biodiversity hotspots

Multiple **estimates of global species richness and endemicity** for different groups identify **regions with higher and unique biodiversity**.

e.g., *the NE Pacific biodiversity hotspot of marine forests of macroalgae (> 50 species locally).*



Fragkopoulou et al, 2022

Unpacking the Ecological Niche

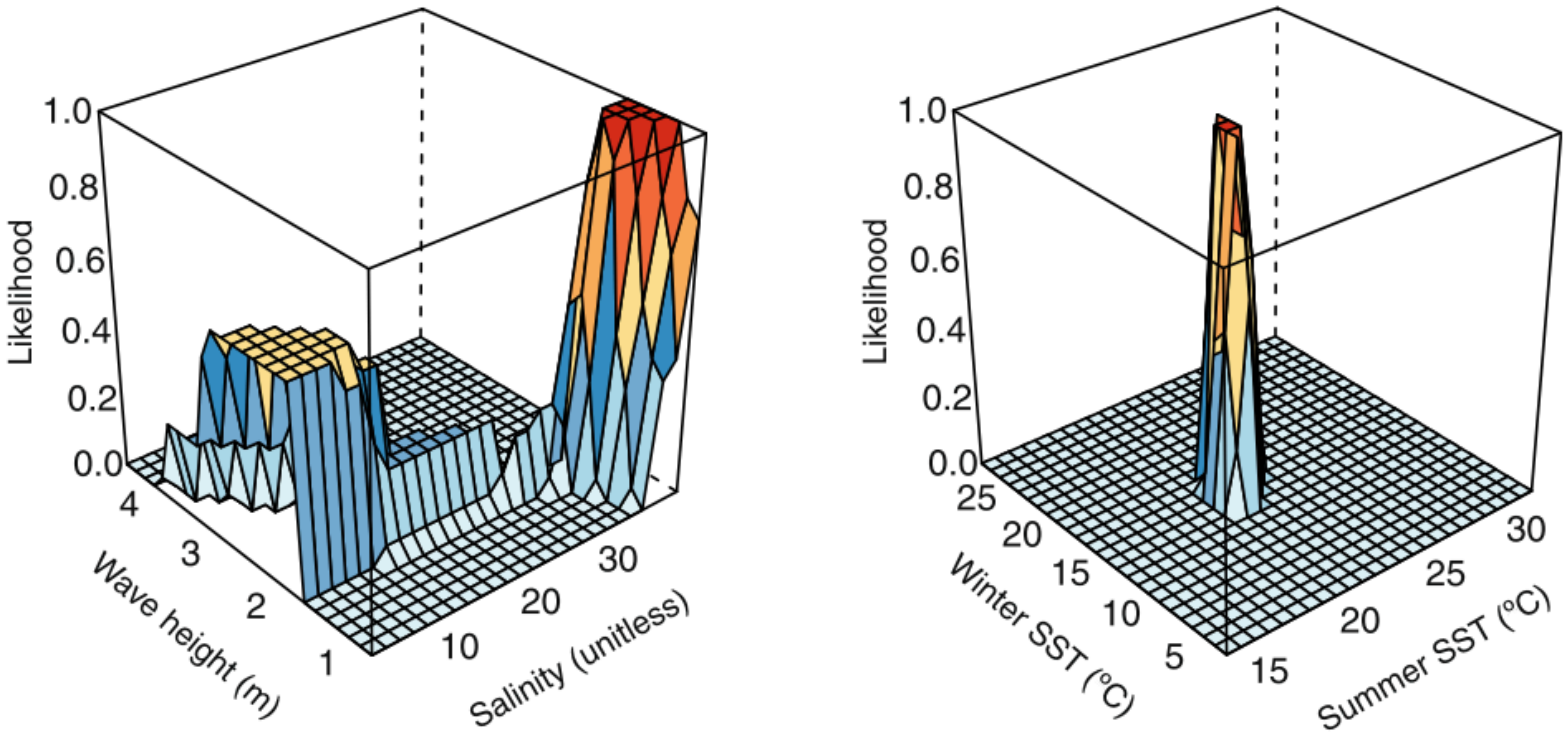
The ecological niche: an evolving core concept

A foundational concept in Macroecology. Evolved through different perspectives:

Grinnellian niche (Joseph Grinnell, 1924): the abiotic conditions (e.g., temperature tolerance) and habitat requirements (e.g., substratum or type) that determine where a species can survive and reproduce.

Eltonian niche (Charles Elton, 1927): the species' functional role within its community, in particularly the interactions it has with other organisms (e.g., competition, predation).

Hutchinsonian niche (G. Evelyn Hutchinson, 1957): a **"n-dimensional hypervolume"** that includes the **abiotic** conditions (i.e., environmental), the **biotic** interactions (e.g., availability of prey) and **resource** requirements (e.g., availability of nursing sites) under which **a species can maintain viable populations** (i.e., survival and reproduction).



Hypervolume of abiotic conditions (environmental dimensions) where a Mediterranean coral can survive and reproduce (fundamental niche).

Boavida et al, 2016

Fundamental vs. realized niche

Hutchinson also distinguished between:

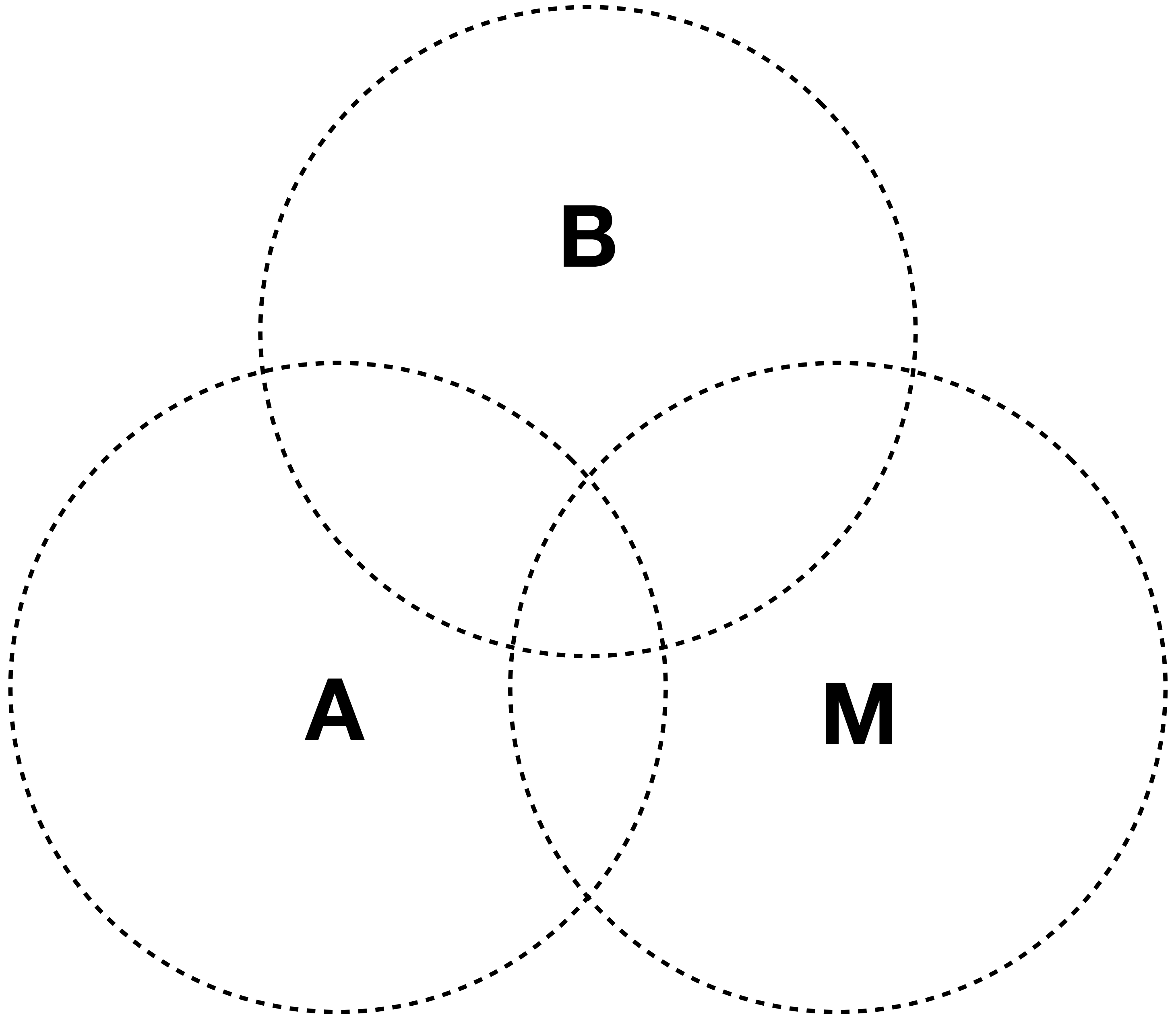
Fundamental niche: the **full range of environmental conditions and resources** (dimensions in the hypervolume) allowing a species to potentially survive and reproduce indefinitely, **in the absence of biotic interactions**.

Realized niche: the **portion of the fundamental niche** (a subset of the hypervolume dimensions) **that a species actually occupies in the presence of limiting biotic interactions** - competition with other species for resources, or predation pressure restricts a species from utilizing its full fundamental niche.

The fundamental niche is the theoretical maximum niche space the species can occupy.

Fundamental Niche > Realized Niche

Connecting Niche to Geography



Connecting niche to geographic space: BAM diagram

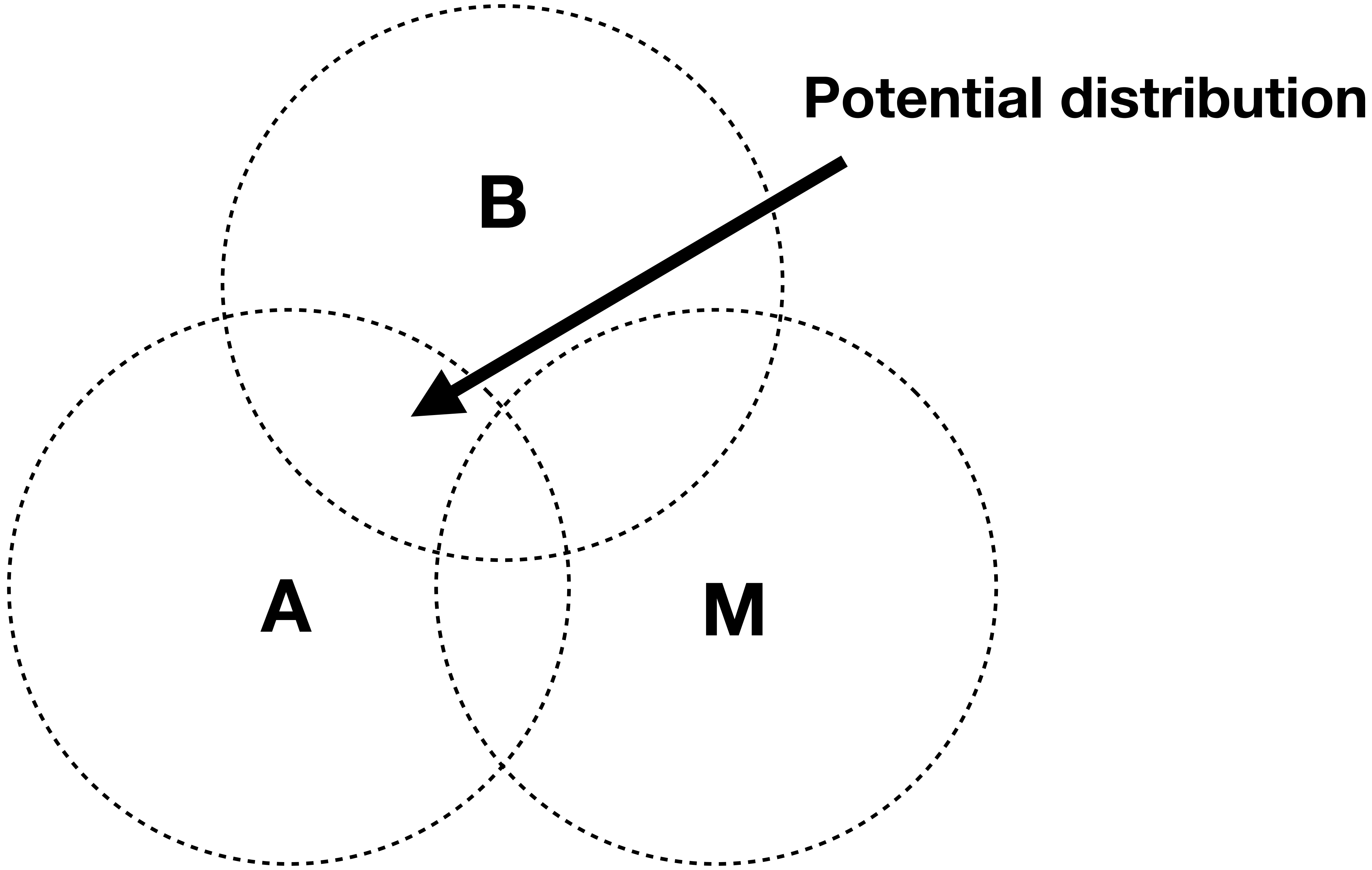
A conceptual model translating niche theory into geographic space. Highlights **three intersecting factors determining distributions:**

Biotic: suitable biotic conditions (presence of prey, absence of strong competitors).

Abiotic: suitable environmental conditions (temperature, salinity, depth, etc.).

Movement: accessibility – the regions reachable by the species via dispersal.

Soberón & Peterson, 2005

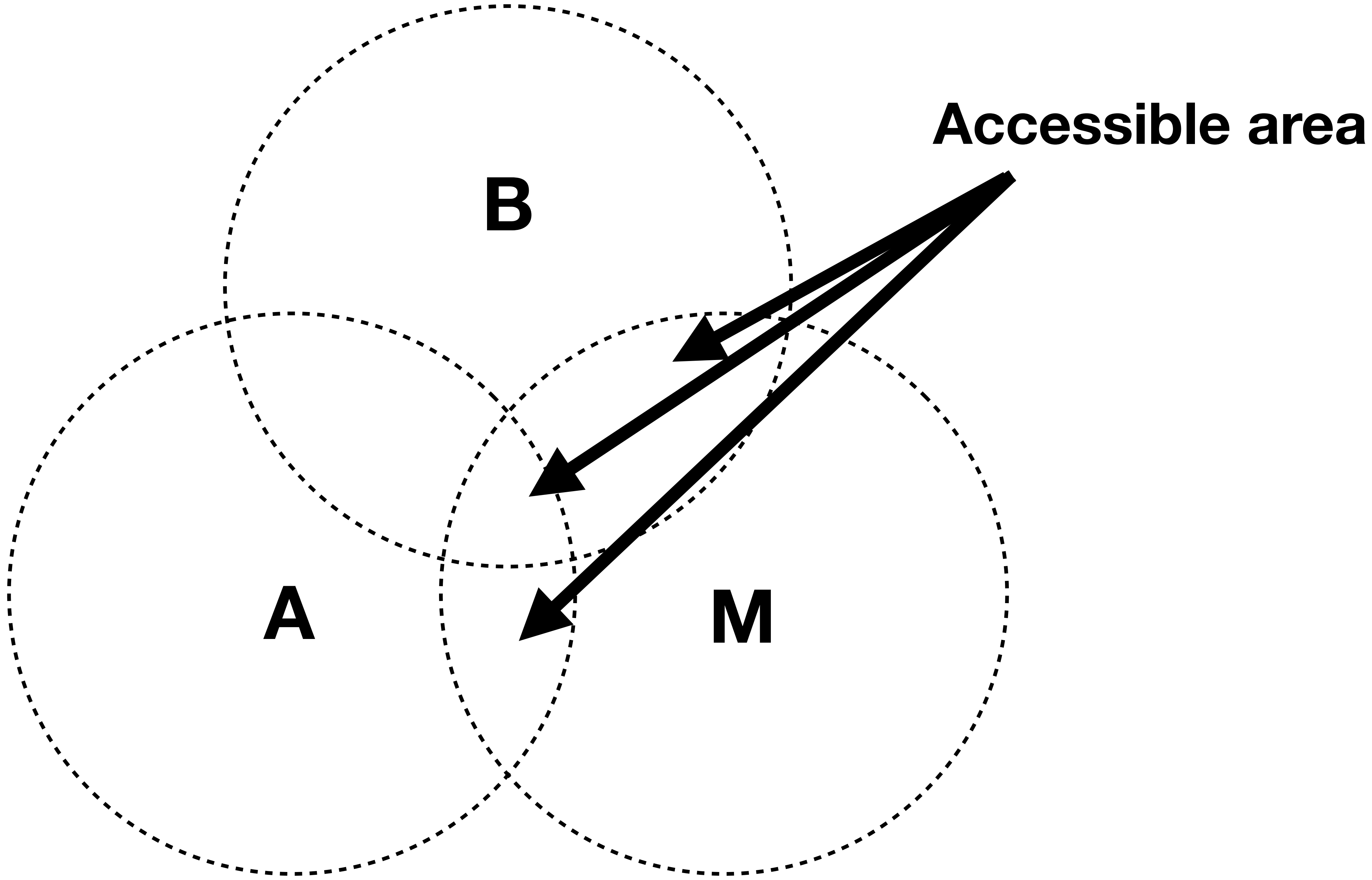


Potential distribution (A ∩ B): The geographic area where both **abiotic (A) and biotic (B) conditions are suitable**.

Accessible area (M): This is the **geographic area the species can actually reach through dispersal** within a given time frame.

Occupied distribution (A ∩ B ∩ M): The **geographic area where the species is actually found**. It's the part of the potential distribution that is also accessible.

Soberón & Peterson, 2005



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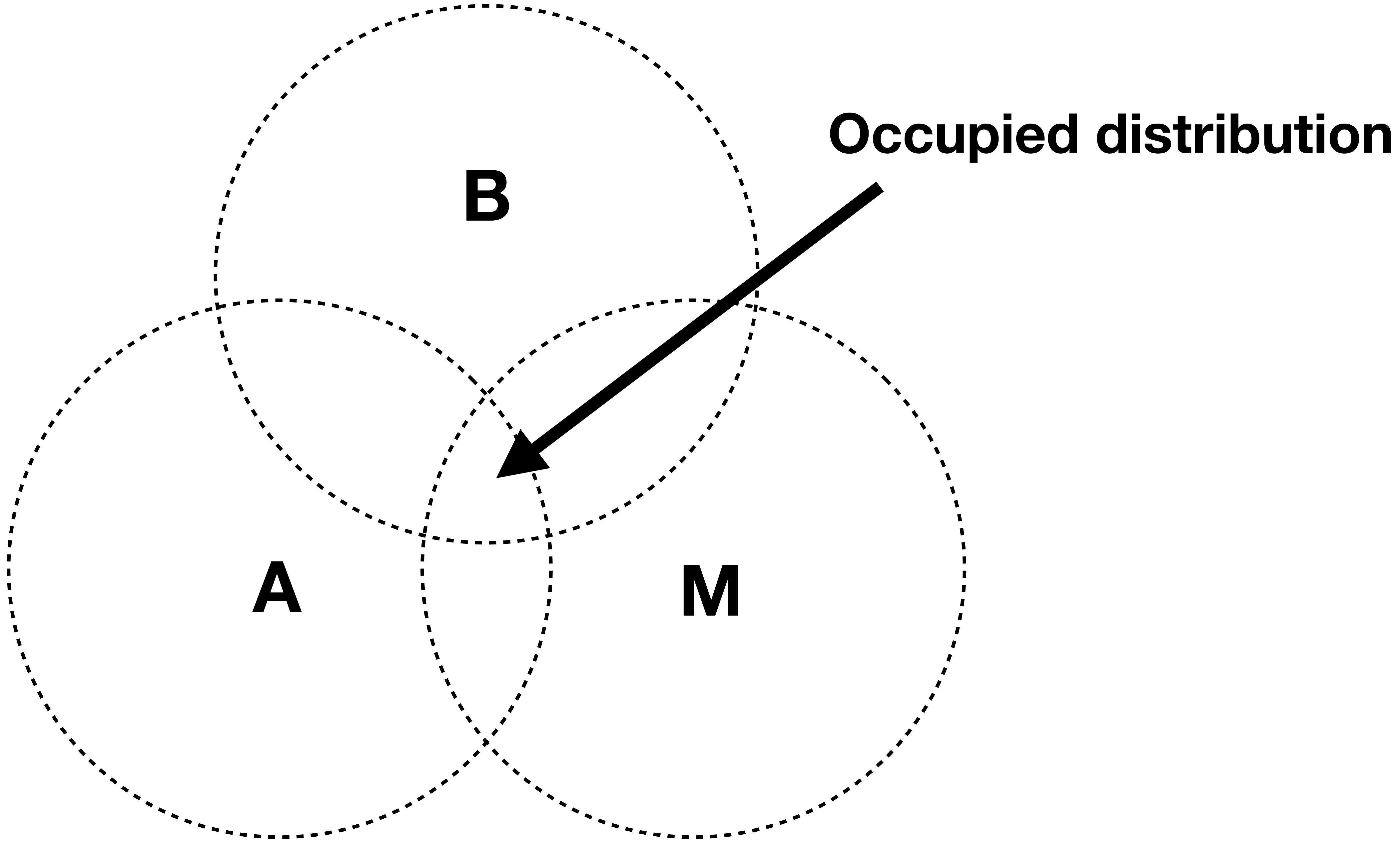

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Are niches fixed? Conservatism vs. evolution

A central question in Macroecology: How stable are species' ecological niches?

Two opposing perspectives:

Niche conservatism: niches remain stable over time.

Niche evolution: niches can evolve over time.

The balance between these perspectives influences species’ distribution, their local adaptation, and speciation.

Niche conservatism

The tendency for **species to retain ancestral niche characteristics** over long periods (e.g., thermal tolerance).

Implications:

- Implies **no adaptive or evolution potential** to novel conditions (fixed or conserved).
- Species **respond to climate change by going extinct or by shifting their geographic distribution** (tracking suitable conditions), rather than adapting in place.

Evidence: observed distribution shifts match climate shifts. Without niche conservatism, species would persist locally while climate conditions shift.

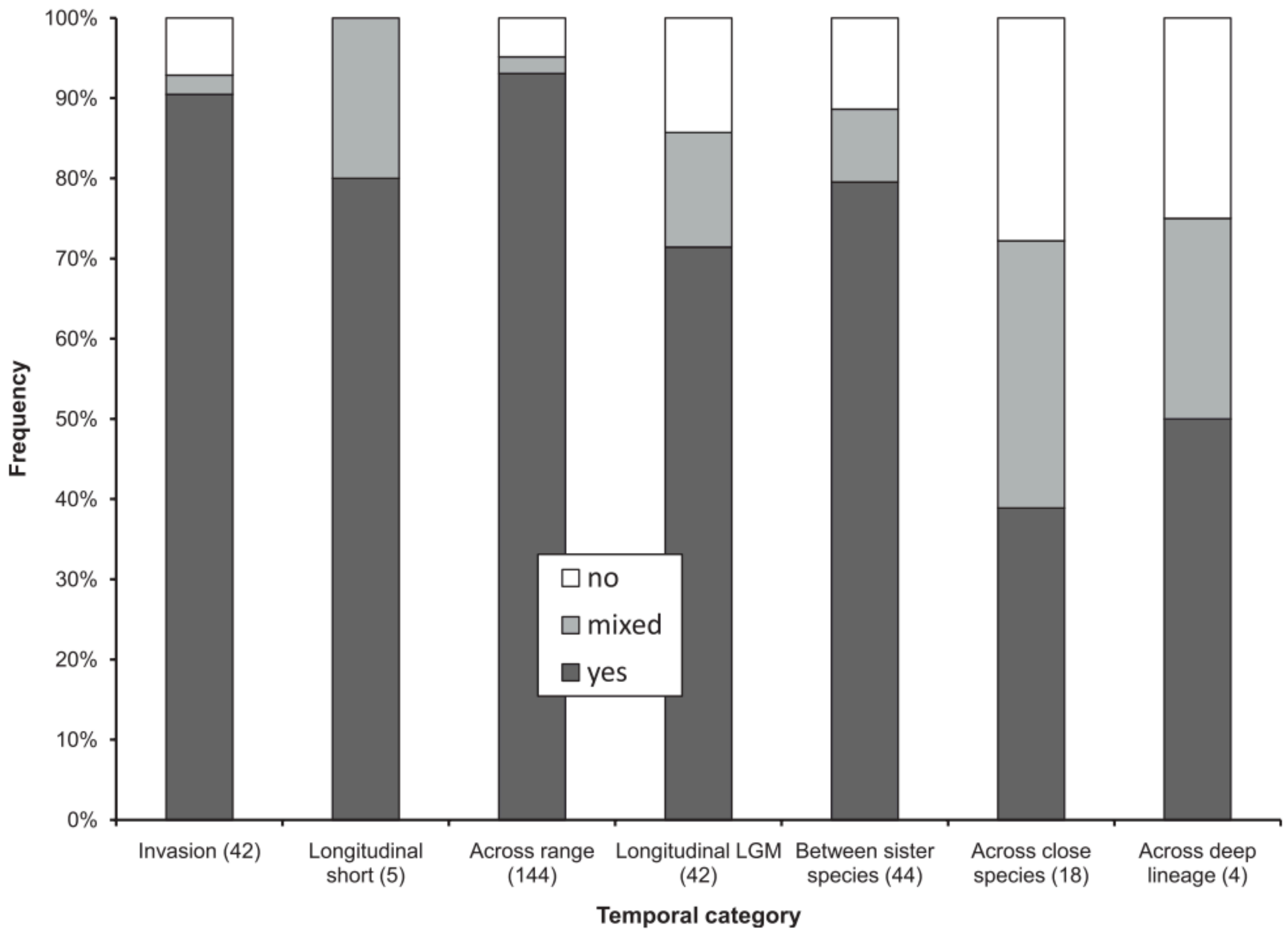
Niche evolution

Niches can evolve, allowing species to colonize new environments.

Happens through natural selection acting on physiological tolerances, resource use, etc.

The key debate today: is niche evolution (i.e., adaptation potential) fast enough to keep the pace of the rapid, human-induced environmental change?

- If evolution is slow relative to climate change, niche conservatism dominates responses (e.g., local extinctions and range shifts).
- If fast enough, adaptation is possible under ongoing climate change (e.g., persistence).



An extensive review **shows evidence for ecological niches being highly conserved from short to deep time spans.**

(i.e., from individual life spans - invasions - up to hundreds of thousands of years).

Even at deep geological time scales of thousands of years NC of ~50% of species.

Peterson et al., 2011

Introduction to Species Distribution Modeling

What is species distribution modeling?

Also known as Environmental Niche Modelling, Habitat Suitability Modelling, …

Process of **using computer algorithms to estimate species ecological niches and predict the distributions** (occurrence, abundance) **across space and time**.

Purpose:

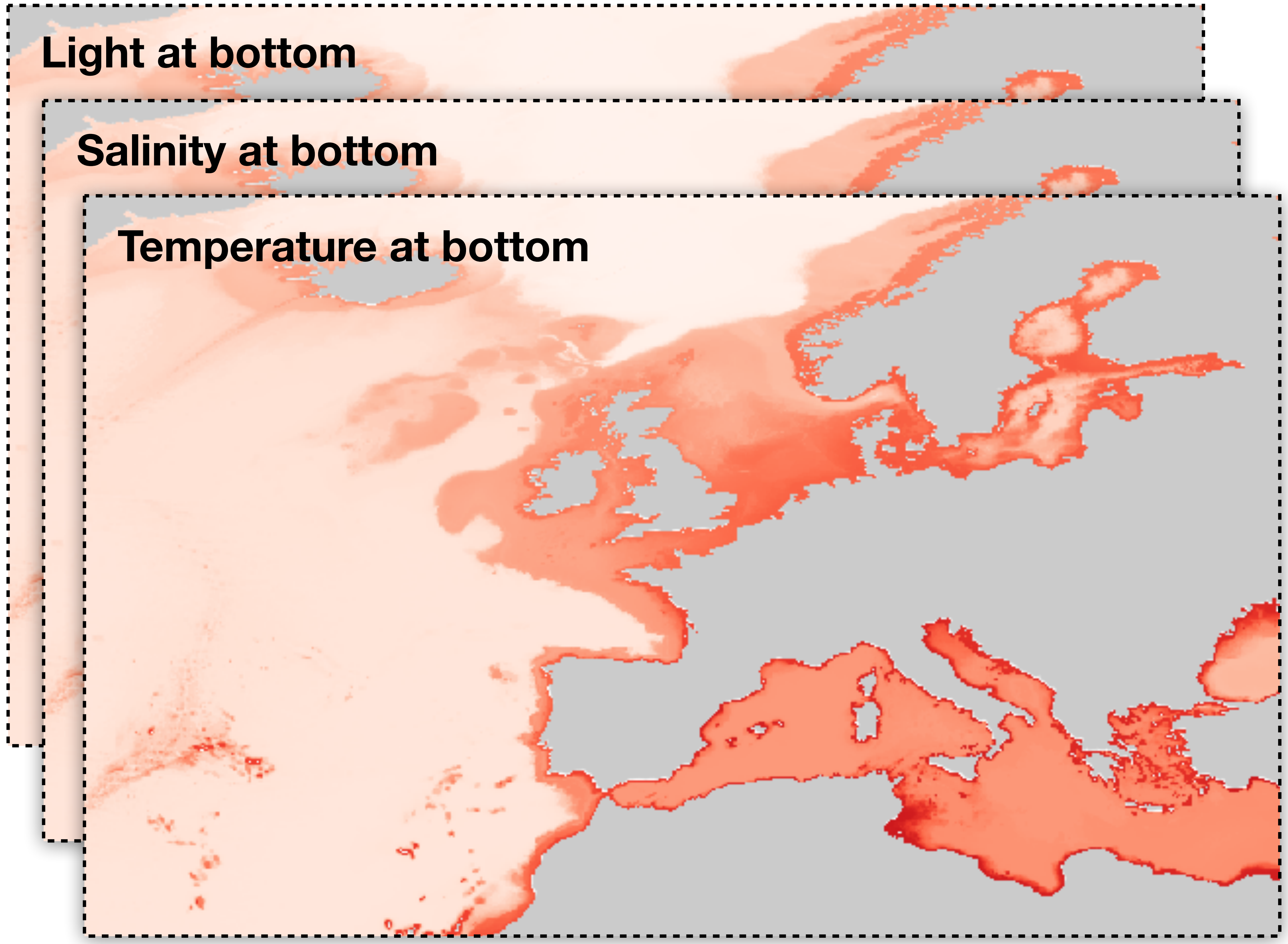
Gain insights into species' environmental tolerances and habitat preferences.

Make spatial predictions of geographic distributions.

Main SDM approaches

Mechanistic modeling: uses **information about the physiological response of species** to environmental conditions (e.g., temperature limits for survival). Require empirical data (often from lab experiments), which may not always be available.

Correlative modeling: based on the **statistical correlation between species occurrence records** (presence-absence) **and the environment**.

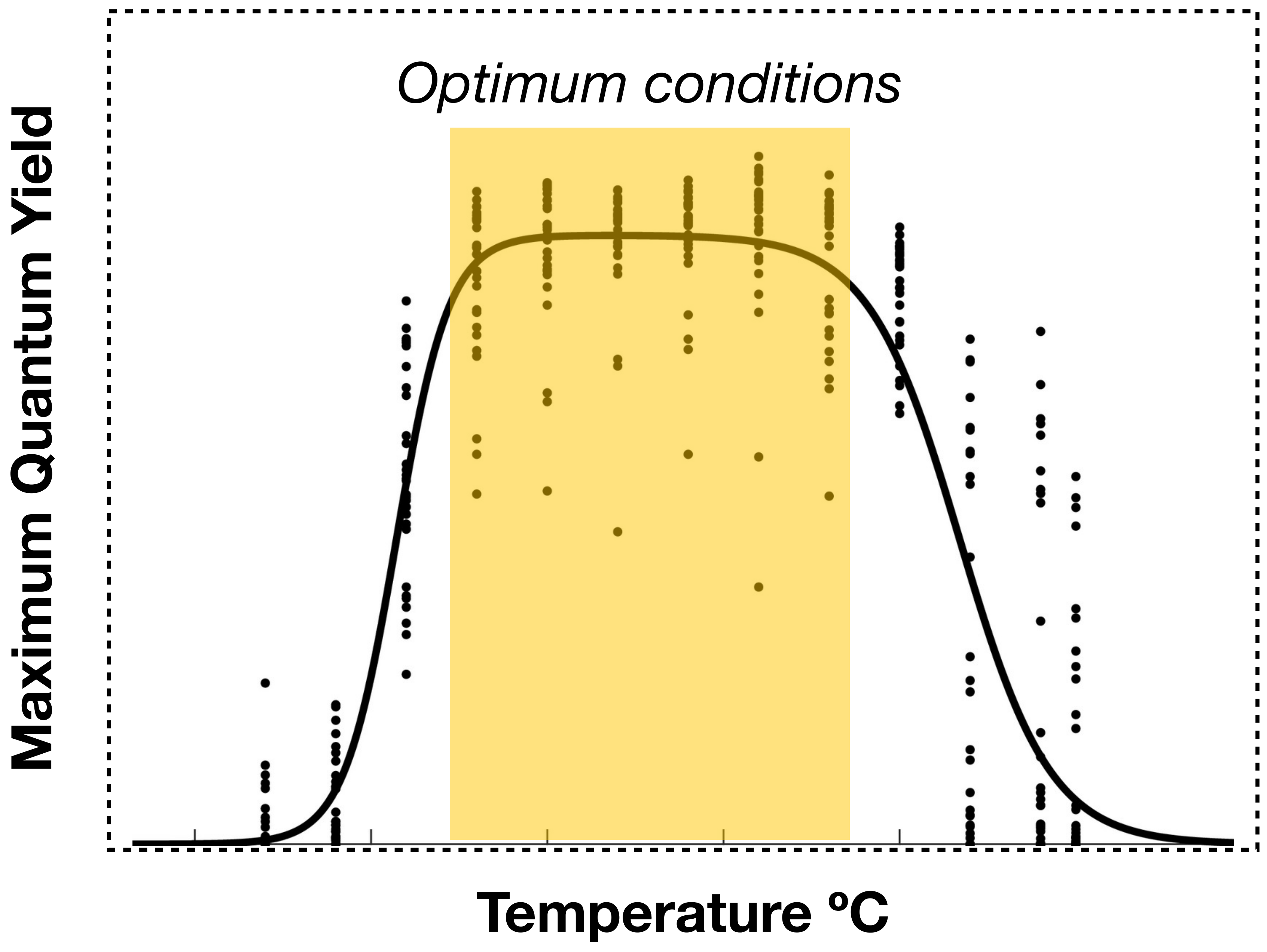


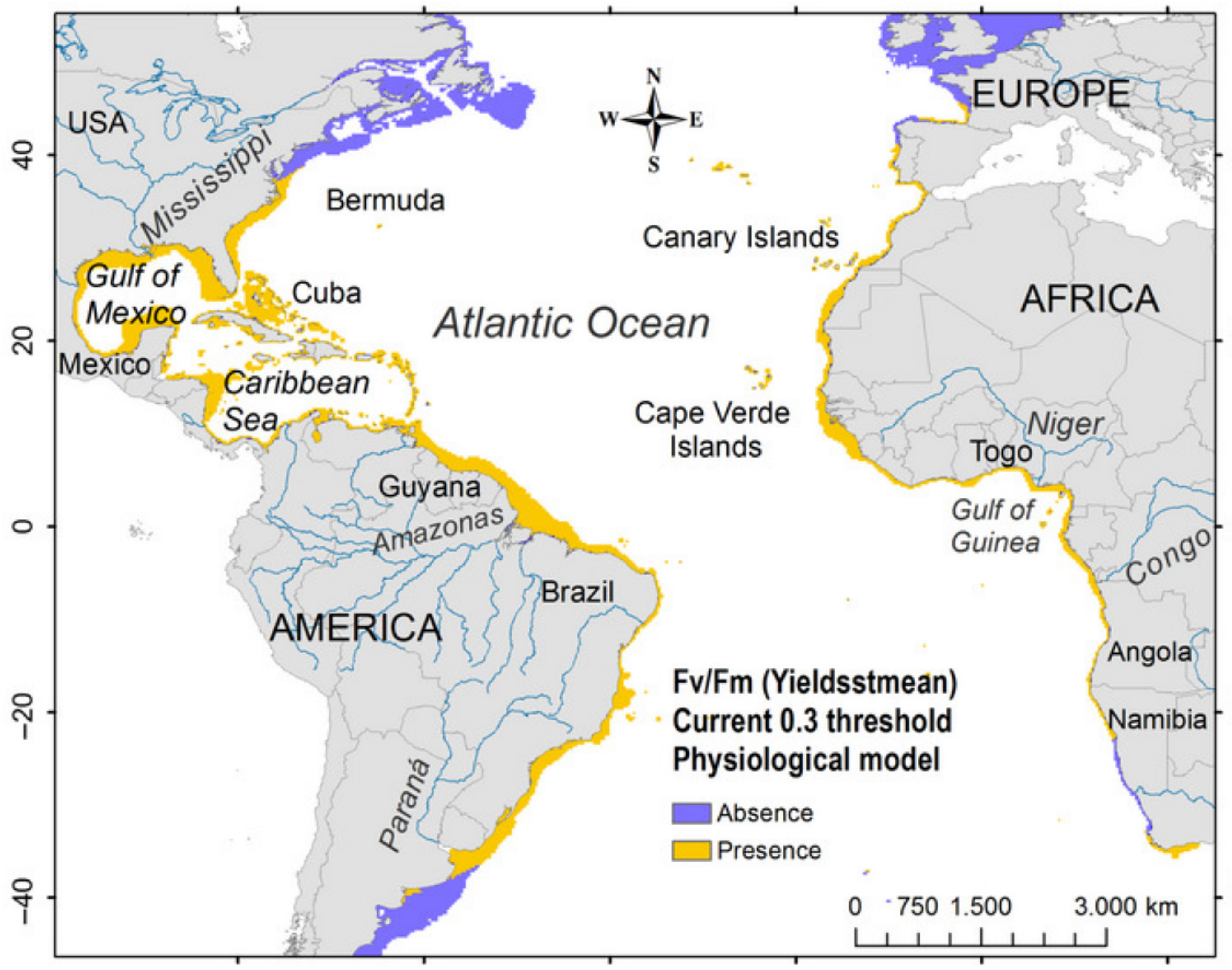
Mechanistic distribution models

Process based: built by reclassifying environmental gradients with tolerance limits.

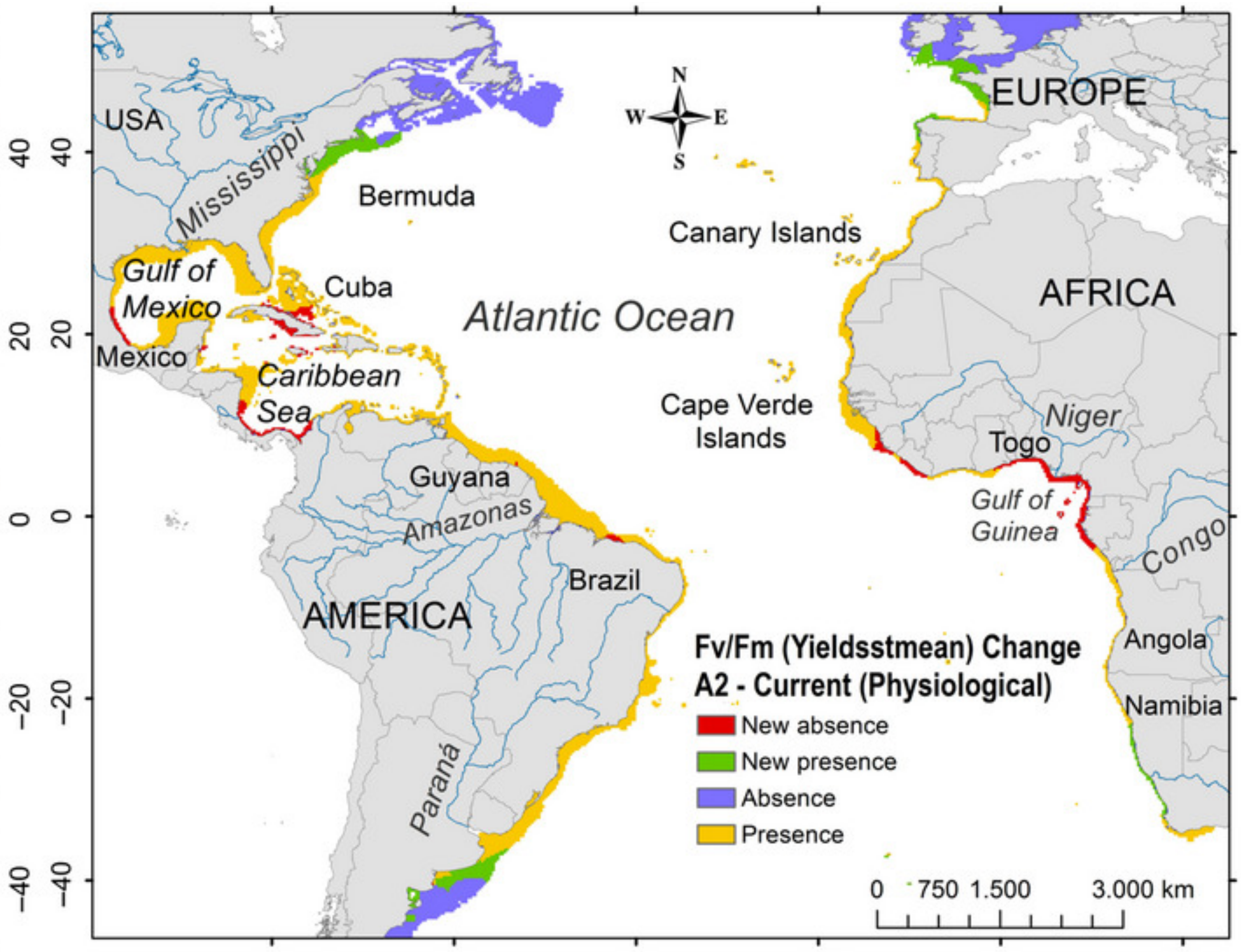
A straightforward approach to predict the distribution of species.

Presence = [Light > 5 $E.m^2.year^{-1}$] ∩ [5 <= Temp. (ºC) <= 20.5ºC] ∩ [Sal. > 28]





Present conditions



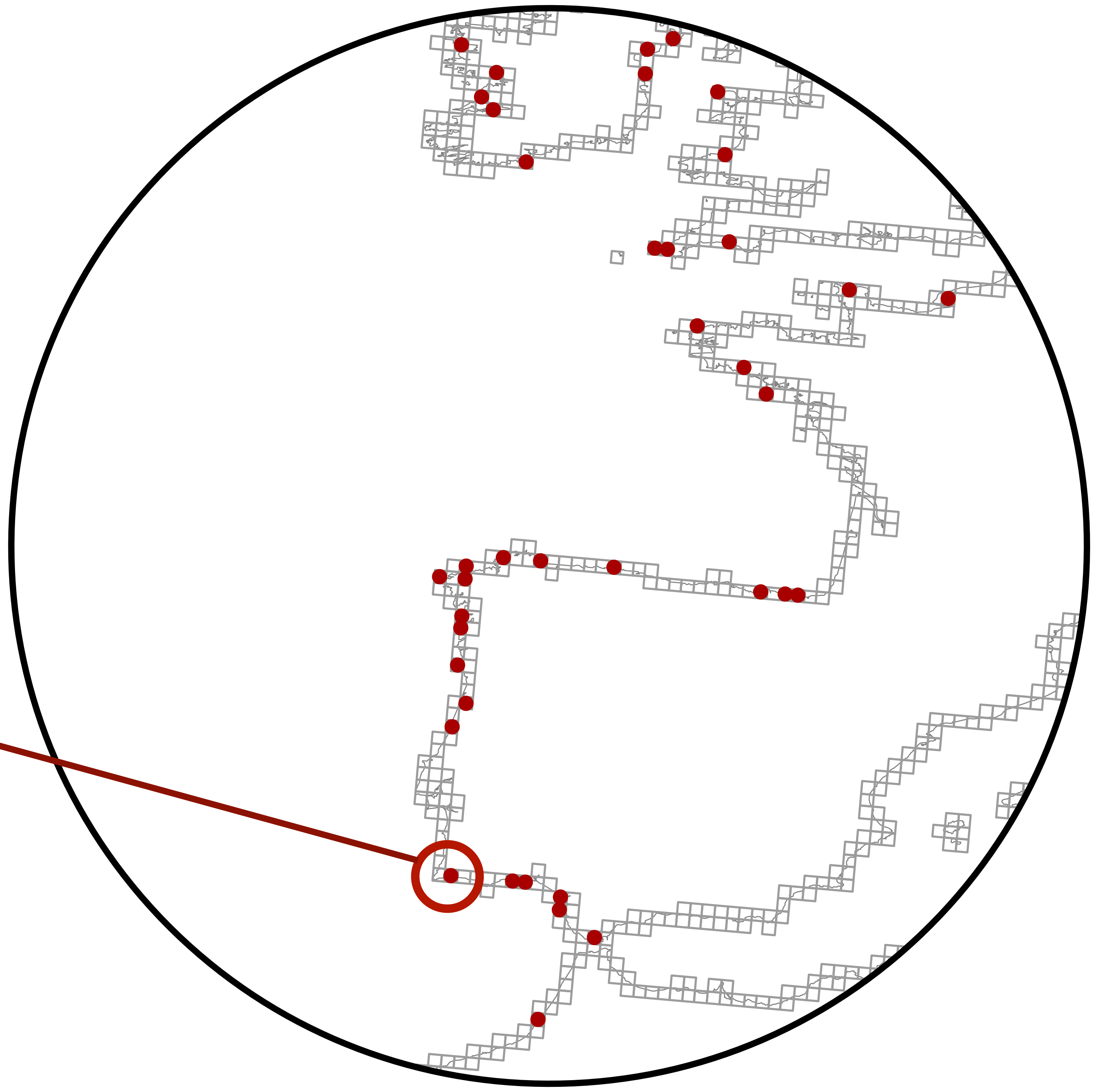
Future conditions

Mechanistic distribution models

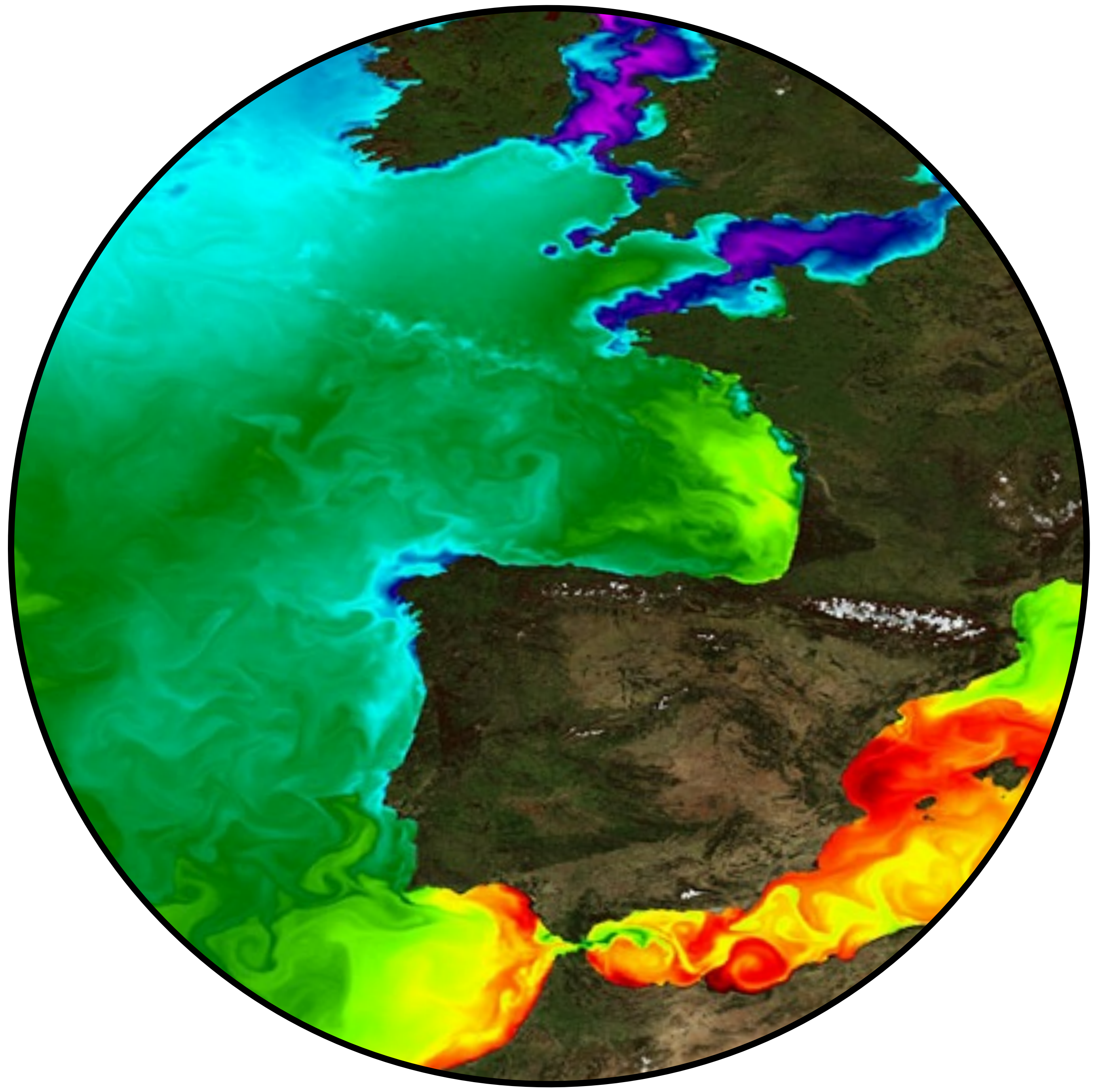
Data driven mechanistic distribution models: (more realistic) by **using response curves from experiments** determining the tolerance of species - from cold and warm conditions (e.g., a species' photosynthetic performance measured by maximal quantum yield) - the optimal conditions can be mapped.

Rodríguez et al., 2019

Obs.	SST	Salinity	Oxygen
1	21.0	35.4	210.1
1	19.4	29.1	224.9
1	21.5	31.1	199.8
0	21.3	33.4	201.1
0	14.8	34.1	184.8
(…)	(…)	(…)	(…)



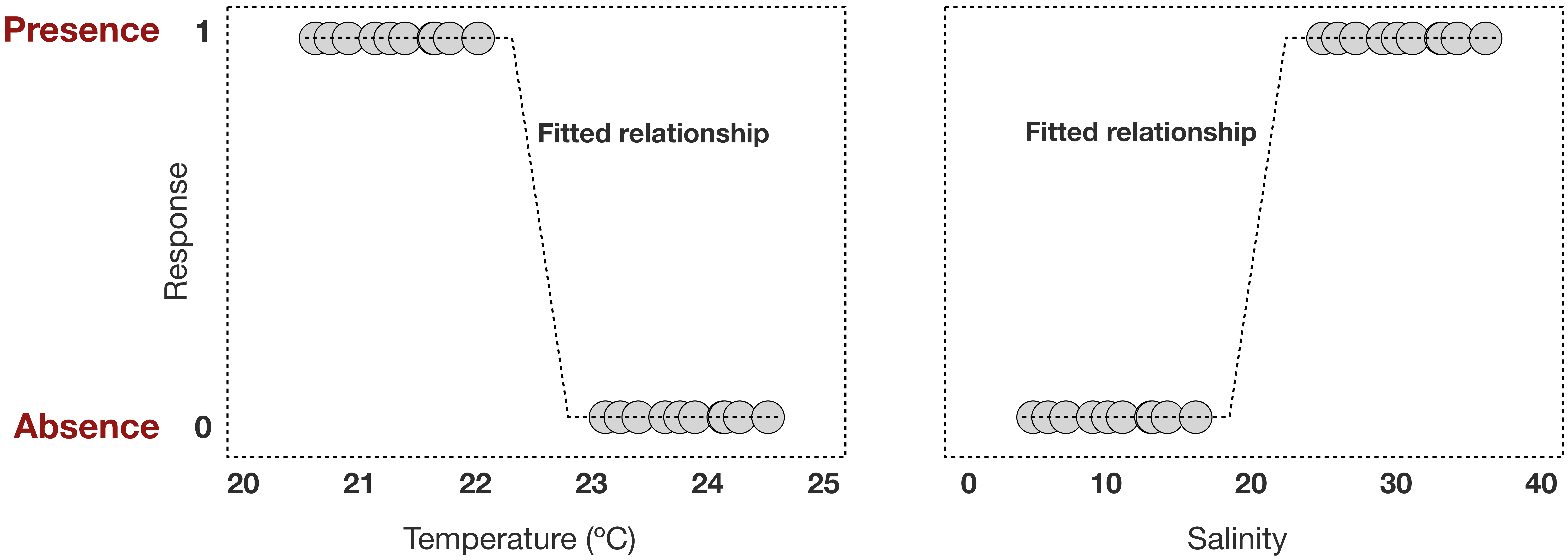
Records of species distribution



Environmental data

Correlative distribution models

Use **statistical correlations between observed species occurrence records and the environmental conditions** at those locations.



The **modeling algorithm learns the effect of environmental** variables **on the distribution of species** (**training data**) and **fits a relationship**.

Connecting niche theory to SDM practice

Correlative SDMs typically model an 'A'-based niche (Grinnellian), thus capturing the abiotic drivers of distributions, and assuming or ignoring B and M.

Under the core assumptions: (1) the observed **geographic distribution is an indicator of a species' environmental requirements**; (2) **niche conservatism** (e.g., to be able to predict future distributions under climate change).

Terrestrial vs. Marine Niche Modeling

Same technical approaches and assumptions. Minor detail changes.

Dimensionality: Terrestrial models often utilize 2D surfaces (elevation, air temp). Marine models must account for a 3D environment (surface vs. benthic depths).

Dispersal: Terrestrial mostly relies on dispersal across static landscapes; marine relies heavily on passive transport via fluid, dynamic ocean currents.

Detectability & bias: Marine environments (especially the deep sea) suffer from severe sampling biases and inaccessibility, leading to poor occurrence data compared to the terrestrial realm.

References

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